Hai Zhou

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Research Interests: Verifiable LLM Reasoning; Semantic Parsing; Knowledge Graphs; Multimodal AI

EDUCATION

University College London *Sep 2025 – Sep 2026*
M.Sc. Integrated Machine Learning Systems London, UK
Current weighted average: 78.06/100 (Distinction-level); final classification pending.
Selected marks: Applied Machine Learning Systems I/II (84/85); Deep Learning for NLP (85).

Nanjing University *Sep 2021 – Jul 2025*
B.Eng. Microelectronic Science and Engineering Nanjing, China
GPA: 4.36/5.0; average score: 87.2/100.
Undergraduate thesis: VO_2-based terahertz reconfigurable leaky-wave antenna (93/100).

RESEARCH EXPERIENCE

Auditable Compositional Question Answering over UK Public Procurement *May 2026 – Present*
M.Sc. Thesis and manuscript in preparation; supervised by Prof. Miguel Rodrigues *University College London*

- **Data and benchmark:** Canonicalised 166,277 UK procurement OCDS releases into a typed knowledge graph with more than 2 million edges, formalising entity identity, monetary additivity, deduplication and record-level provenance; built a 12,828-question executable benchmark spanning retrieval, aggregation, comparison, multi-hop composition and typed abstention, with 99.88% agreement between independently implemented oracles.
- **System and training:** Developed a grammar-constrained LLM planner coupled to deterministic grounding, compilation, execution and evidence verification, which remained authoritative for every final answer; fine-tuned Qwen3-8B and Llama-3.1-8B LoRA student planners exclusively on execution-verified candidate traces.
- **Validation and transfer:** Achieved 85.65% strict accuracy on a sealed 2,285-question procurement test and 78.31% on the 922-question PACS compositional challenge set, surpassing the cloud teacher that generated the training candidates by 15.9 and 28.0 percentage points, respectively; transferred the architecture to WikiTableQuestions by reusing all 17 original operators with one table-specific extension, where gold-program supervision reached 51.80% on the official test.

Multimodal Document Reranking with Vision-Language Models *Jan 2026 – Mar 2026*
Research Project *University College London*

- Improved MMDocIR Macro Recall@1 from 0.648 to 0.695 over the ColQwen2 retriever with a retriever-agnostic vision-language reranker that jointly reasons over text, tables, charts and page layout; built 7,200 rationale-supervised examples and fine-tuned Qwen2.5-VL-7B through a two-stage SFT–RL pipeline.

SELECTED RESEARCH PROJECTS

KG-Grounded Preference Optimisation for Multi-Hop Question Answering *Apr 2026 – Jul 2026*
Course Research Project, Deep Learning for NLP *University College London*

- Constructed 387 preference pairs without human annotation by fusing R-GCN structural plausibility with NLI entailment, then LoRA-tuned Qwen2-0.5B with a reference-free DPO objective using no KG access at inference; showed that structural and textual grounding metrics diverge in entity-centric multi-hop KGQA.

GUI Element Localisation with Vision-Language Models *Sep 2025 – Nov 2025*
Course Research Project *University College London*

- Built a DOM-independent visual-grounding model that predicts click coordinates directly from desktop, web and mobile screenshots; reused each screenshot across 30 element queries for 30× denser supervision and fine-tuned Qwen2.5-VL-3B with LoRA to 84.9% point-in-bounding-box accuracy on ScreenSpot, over 93% on text elements.

TECHNICAL SKILLS

Programming and Systems: Python, SQL, JavaScript/TypeScript, Linux, Git, Docker, Pandas, Parquet.
ML and LLM Systems: PyTorch, Transformers, PEFT/LoRA, TRL, vLLM, SFT, preference optimisation, constrained decoding, vision-language models, knowledge graphs.

HONOURS & AWARDS

National Second Prize, Chinese Collegiate Computing Competition *2023*
Honorable Mention, Mathematical Contest in Modeling (MCM) *2024*
Provincial Innovation & Entrepreneurship Training Grant *2023 – 2025*
Renmin Scholarship, Nanjing University – awarded for three consecutive years *2022 – 2024*

LEADERSHIP & SERVICE

Captain, Electronics Debate Team, Nanjing University; 2024 School Cup champions *2022 – 2024*